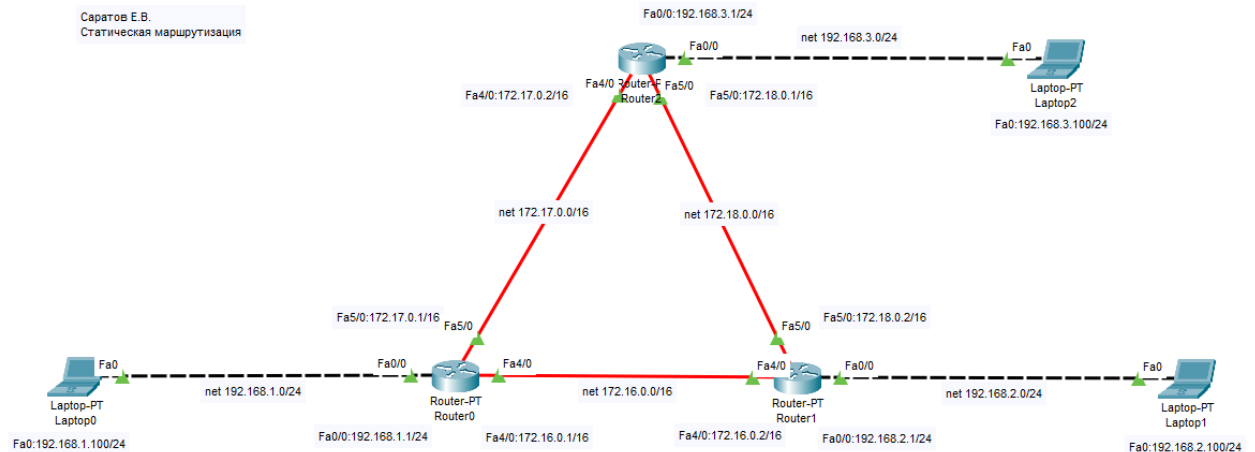


Практическая работа

Сеть со статической маршрутизацией



Трассировка

с Laptop0 — трассировку 192.168.2.100

```
Laptop0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.2.100

Tracing route to 192.168.2.100 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.1.1
  2  *        *        *        Request timed out.
  3  0 ms    0 ms    0 ms    172.16.0.2
  4  *        0 ms    0 ms    192.168.2.100

Trace complete.

C:\>tracert 192.168.2.100

Tracing route to 192.168.2.100 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.1.1
  2  0 ms    0 ms    0 ms    172.18.0.1
  3  0 ms    0 ms    0 ms    172.16.0.2
  4  0 ms    0 ms    0 ms    192.168.2.100

Trace complete.

C:\>tracert 192.168.2.100

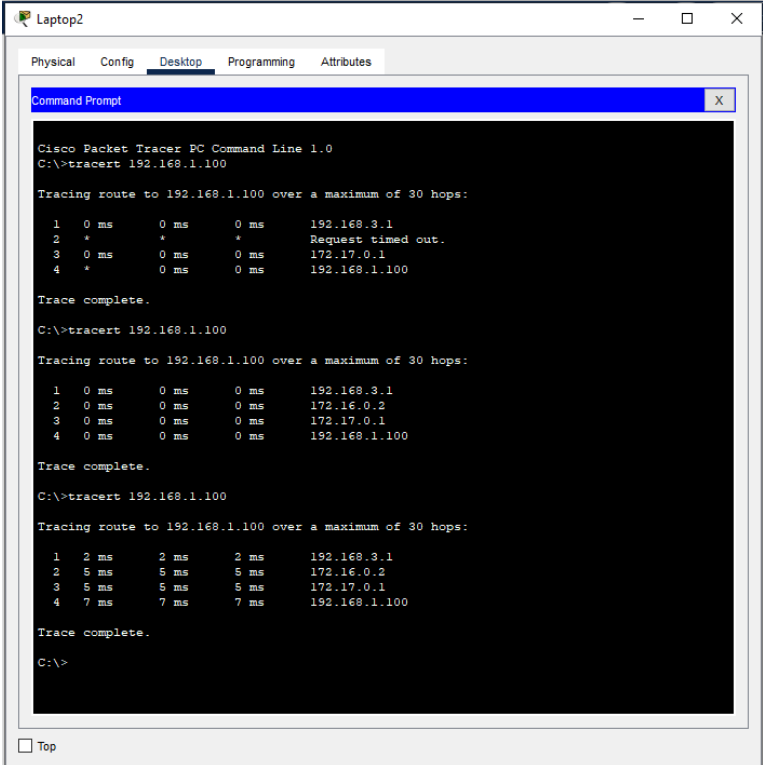
Tracing route to 192.168.2.100 over a maximum of 30 hops:

  1  2 ms    2 ms    2 ms    192.168.1.1
  2  5 ms    5 ms    5 ms    172.18.0.1
  3  5 ms    5 ms    5 ms    172.16.0.2
  4  7 ms    7 ms    7 ms    192.168.2.100

Trace complete.

C:\>
```

Laptop2 — 192.168.1.100



The screenshot shows a Cisco Packet Tracer PC Command Line window for Laptop2. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, showing a Command Prompt window. The Command Prompt displays three successful traceroute commands to 192.168.1.100. The first two show 0 ms delays, while the third shows delays of 2 ms, 5 ms, 5 ms, and 7 ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.1.100

Tracing route to 192.168.1.100 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.3.1
  2  *        *        *        Request timed out.
  3  0 ms    0 ms    0 ms    172.17.0.1
  4  *        0 ms    0 ms    192.168.1.100

Trace complete.

C:\>tracert 192.168.1.100

Tracing route to 192.168.1.100 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.3.1
  2  0 ms    0 ms    0 ms    172.16.0.2
  3  0 ms    0 ms    0 ms    172.17.0.1
  4  0 ms    0 ms    0 ms    192.168.1.100

Trace complete.

C:\>tracert 192.168.1.100

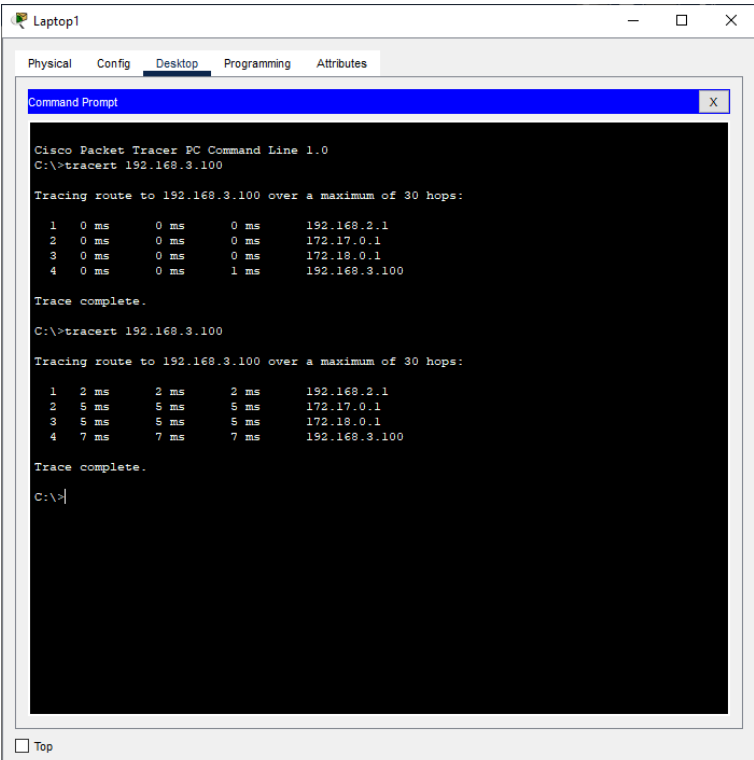
Tracing route to 192.168.1.100 over a maximum of 30 hops:

  1  2 ms    2 ms    2 ms    192.168.3.1
  2  5 ms    5 ms    5 ms    172.16.0.2
  3  5 ms    5 ms    5 ms    172.17.0.1
  4  7 ms    7 ms    7 ms    192.168.1.100

Trace complete.

C:\>
```

Laptop1 — 192.168.3.100



The screenshot shows a Cisco Packet Tracer PC Command Line window for Laptop1. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, showing a Command Prompt window. The Command Prompt displays two successful traceroute commands to 192.168.3.100. The first shows 0 ms delays, and the second shows delays of 2 ms, 5 ms, 5 ms, and 7 ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.3.100

Tracing route to 192.168.3.100 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.2.1
  2  0 ms    0 ms    0 ms    172.17.0.1
  3  0 ms    0 ms    0 ms    172.18.0.1
  4  0 ms    0 ms    1 ms    192.168.3.100

Trace complete.

C:\>tracert 192.168.3.100

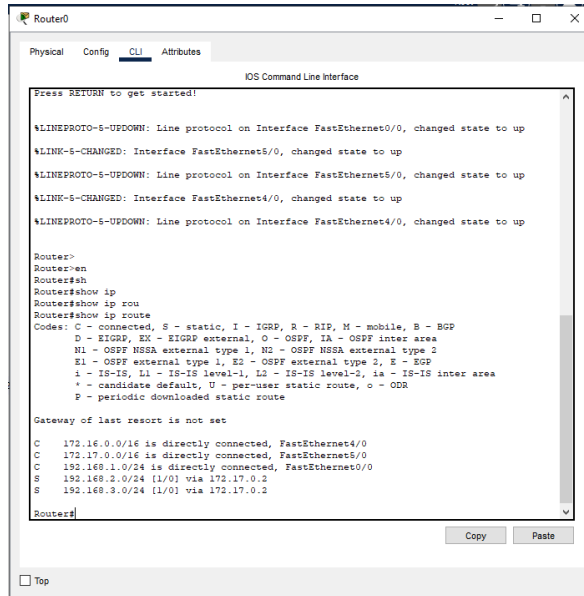
Tracing route to 192.168.3.100 over a maximum of 30 hops:

  1  2 ms    2 ms    2 ms    192.168.2.1
  2  5 ms    5 ms    5 ms    172.17.0.1
  3  5 ms    5 ms    5 ms    172.18.0.1
  4  7 ms    7 ms    7 ms    192.168.3.100

Trace complete.

C:\>
```

Выполнения команды sh ip ro

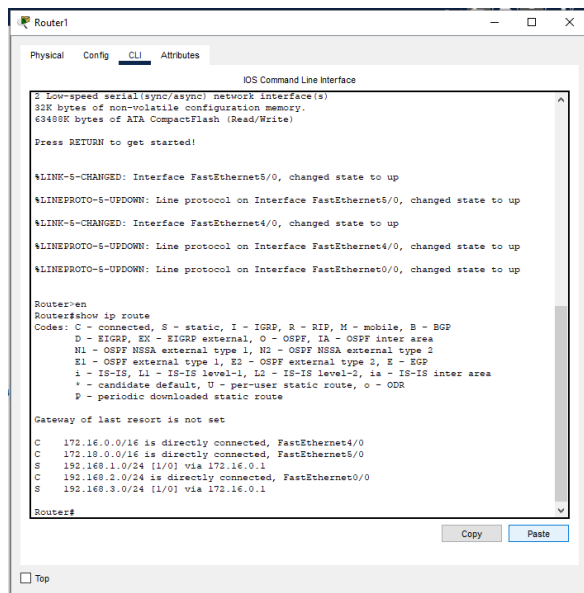


```
Router0>
Router0>en
Router0#sh
Router0#show ip
Router0#show ip rou
Router0#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C    172.16.0.0/16 is directly connected, FastEthernet0/0
C    172.17.0.0/16 is directly connected, FastEthernet5/0
C    192.168.1.0/24 is directly connected, FastEthernet0/0
S    192.168.2.0/24 [1/0] via 172.17.0.2
S    192.168.3.0/24 [1/0] via 172.17.0.2

Router0#
```

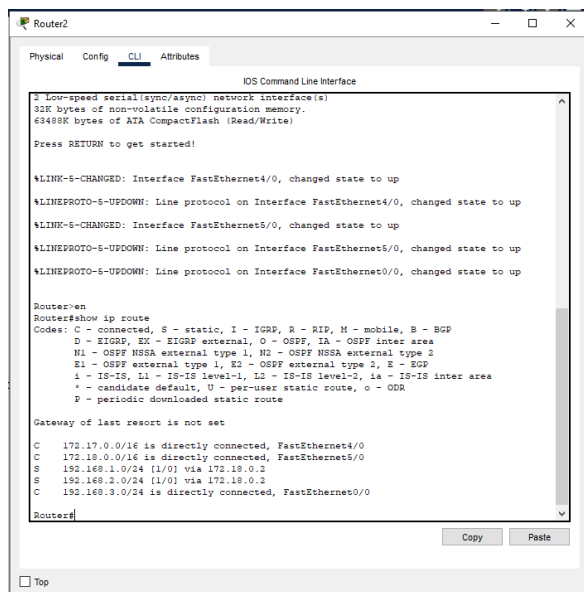


```
Router1>
Router1>en
Router1#sh
Router1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C    172.16.0.0/16 is directly connected, FastEthernet4/0
C    172.18.0.0/16 is directly connected, FastEthernet5/0
S    192.168.1.0/24 [1/0] via 172.16.0.1
S    192.168.2.0/24 is directly connected, FastEthernet0/0
S    192.168.3.0/24 [1/0] via 172.16.0.1

Router1#
```



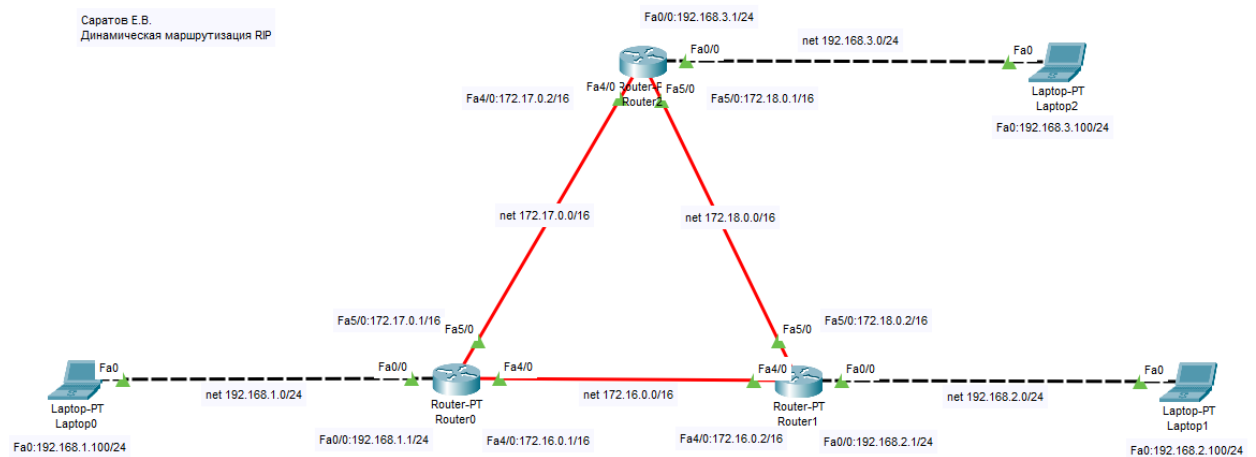
```
Router2>
Router2>en
Router2#sh
Router2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C    172.17.0.0/16 is directly connected, FastEthernet4/0
C    172.18.0.0/16 is directly connected, FastEthernet5/0
S    192.168.1.0/24 [1/0] via 172.18.0.2
S    192.168.2.0/24 [1/0] via 172.18.0.2
C    192.168.3.0/24 is directly connected, FastEthernet0/0

Router2#
```

Динамическая маршрутизация RIP



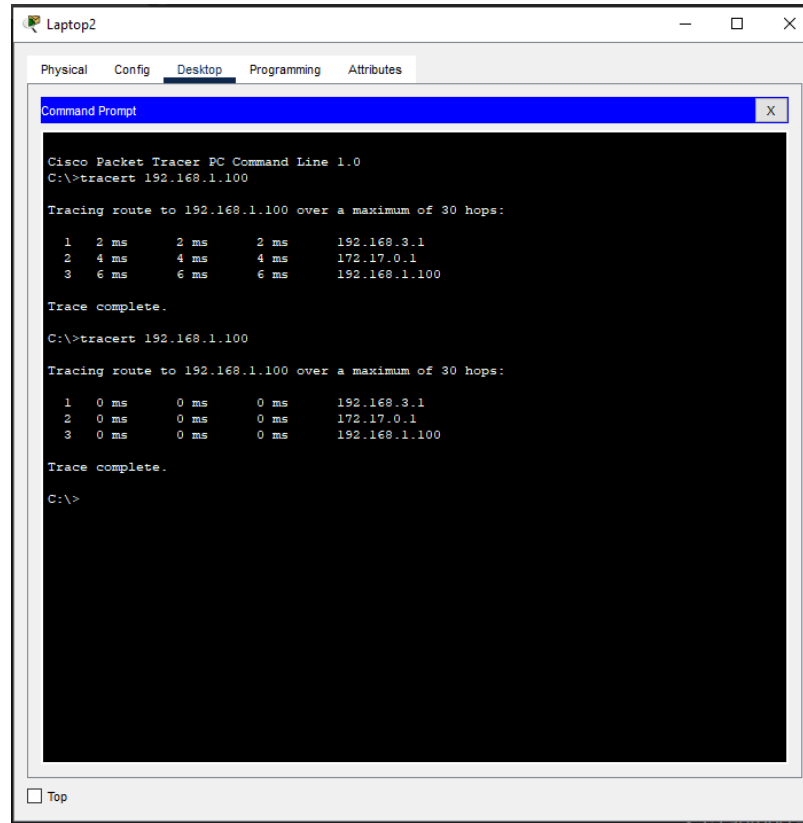
Выбран протокол RIP, т.к. сеть небольшая и проще применить и настроить этот протокол.

Трассировка

с Laptop0 — трассировку 192.168.2.100

```
Laptop0
Physical Config Desktop Programming Attributes
Command Prompt
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
  Minimum = 6ms, Maximum = 6ms, Average = 6ms
C:\>ping 192.168.2.100
Pinging 192.168.2.100 with 32 bytes of data:
Request timed out.
Reply from 192.168.2.100: bytes=32 time=6ms TTL=126
Reply from 192.168.2.100: bytes=32 time=6ms TTL=126
Reply from 192.168.2.100: bytes=32 time=6ms TTL=126
Ping statistics for 192.168.2.100:
  Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
  Minimum = 6ms, Maximum = 6ms, Average = 6ms
C:\>tracert 192.168.2.100
Tracing route to 192.168.2.100 over a maximum of 30 hops:
  0  0 ms  0 ms  0 ms  192.168.1.1
  1  0 ms  0 ms  0 ms  172.16.0.2
  2  0 ms  0 ms  0 ms  192.168.2.100
Trace complete.
C:\>tracert 192.168.2.100
Tracing route to 192.168.2.100 over a maximum of 30 hops:
  0  2 ms  2 ms  2 ms  192.168.1.1
  1  4 ms  4 ms  4 ms  172.16.0.2
  2  6 ms  6 ms  6 ms  192.168.2.100
Trace complete.
C:\>
```

Laptop2 — 192.168.1.100



The screenshot shows the 'Laptop2' window with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the output of two 'tracert' commands. The first command shows a successful trace to 192.168.1.100 with three hops and increasing latency. The second command shows a successful trace to 192.168.1.100 with three hops and zero latency.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.1.100

Tracing route to 192.168.1.100 over a maximum of 30 hops:

  1  2 ms    2 ms    2 ms    192.168.3.1
  2  4 ms    4 ms    4 ms    172.17.0.1
  3  6 ms    6 ms    6 ms    192.168.1.100

Trace complete.

C:\>tracert 192.168.1.100

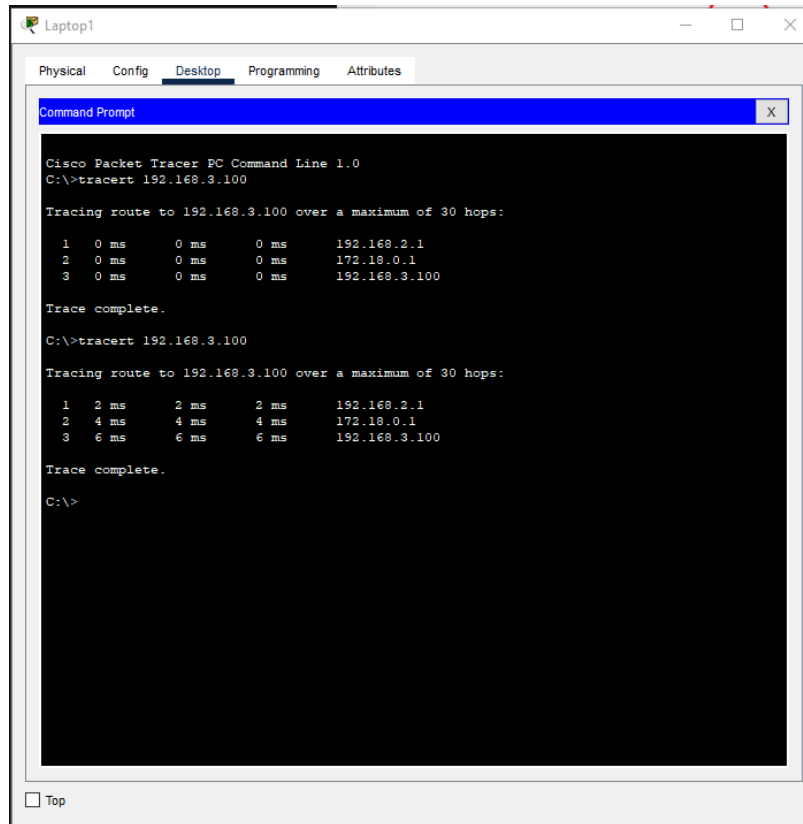
Tracing route to 192.168.1.100 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.3.1
  2  0 ms    0 ms    0 ms    172.17.0.1
  3  0 ms    0 ms    0 ms    192.168.1.100

Trace complete.

C:\>
```

Laptop1 — 192.168.3.100



The screenshot shows the 'Laptop1' window with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the output of two 'tracert' commands. The first command shows a successful trace to 192.168.3.100 with three hops and zero latency. The second command shows a successful trace to 192.168.3.100 with three hops and increasing latency.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.3.100

Tracing route to 192.168.3.100 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.3.1
  2  0 ms    0 ms    0 ms    172.18.0.1
  3  0 ms    0 ms    0 ms    192.168.3.100

Trace complete.

C:\>tracert 192.168.3.100

Tracing route to 192.168.3.100 over a maximum of 30 hops:

  1  2 ms    2 ms    2 ms    192.168.2.1
  2  4 ms    4 ms    4 ms    172.18.0.1
  3  6 ms    6 ms    6 ms    192.168.3.100

Trace complete.

C:\>
```

Выполнения команды sh ip ro

